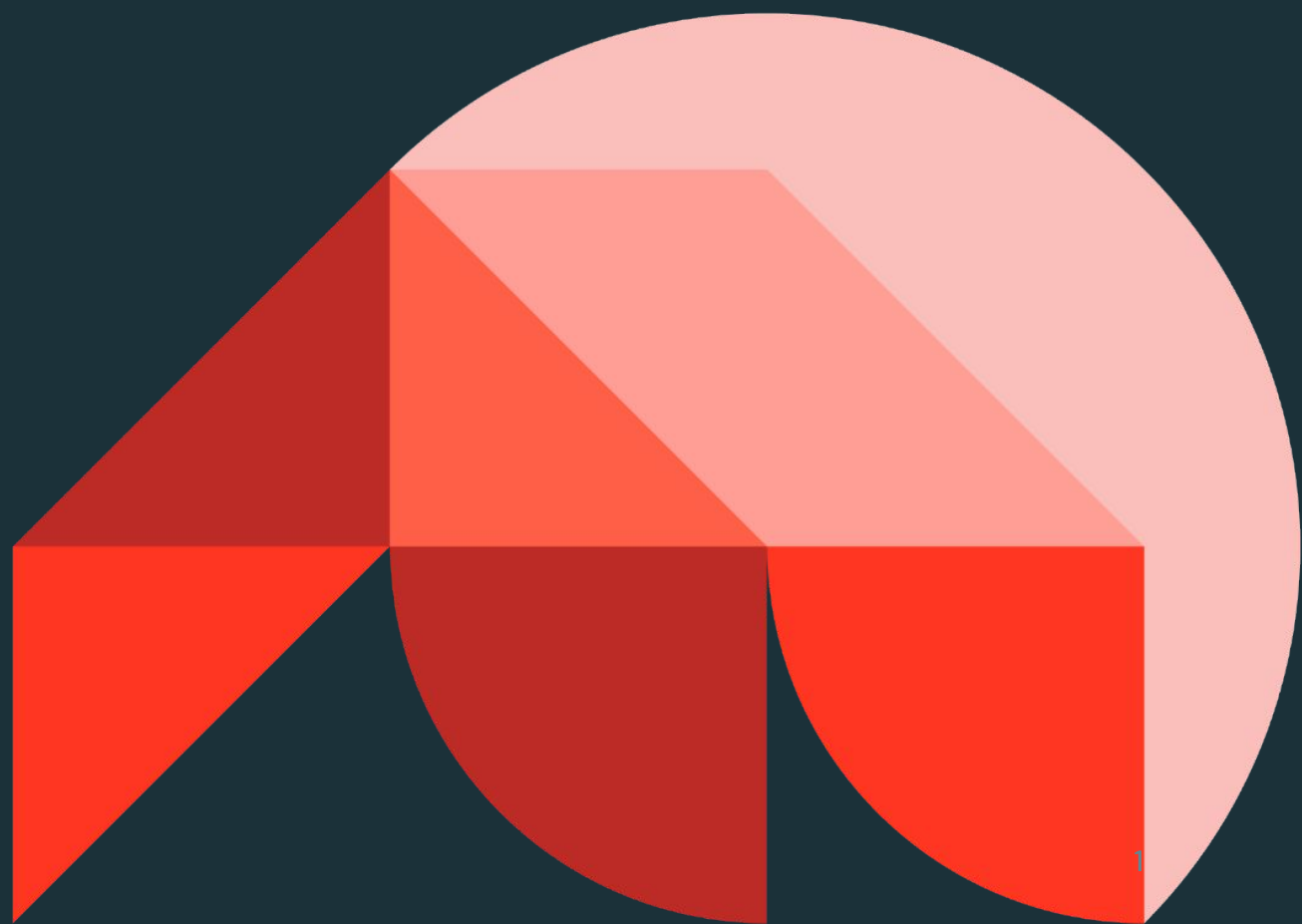




Introduction to Databricks

Dong Qiao Yang



Agenda

- What is Databricks?
- Fundamentals
- Introduction to Databricks Notebooks
- User experience: Ecosystem & ease of use
- Hands On workshop 🎉🎉🎉



What is Databricks?





Agent Bricks
Production AI agents



AI/BI
Agentic business intelligence



Custom Apps
Secure data and AI apps

And more...



AI with Enterprise Context



Genie



Lakehouse
Data warehousing



Lakebase
Serverless Postgres



Lakeflow
Ingest, ETL, streaming



Unified Governance



Unity Catalog



Open Formats



Postgres

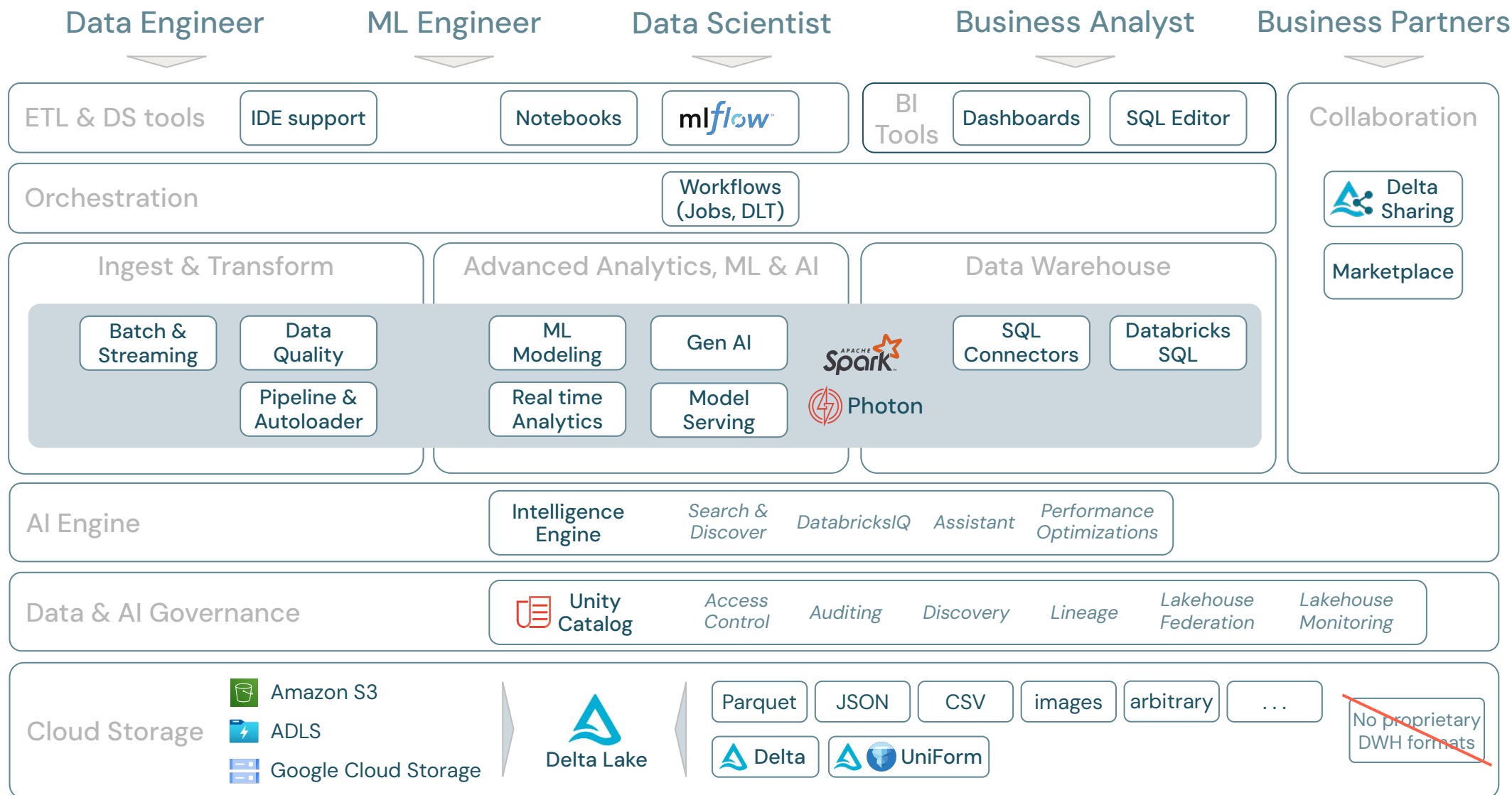


DELTA LAKE

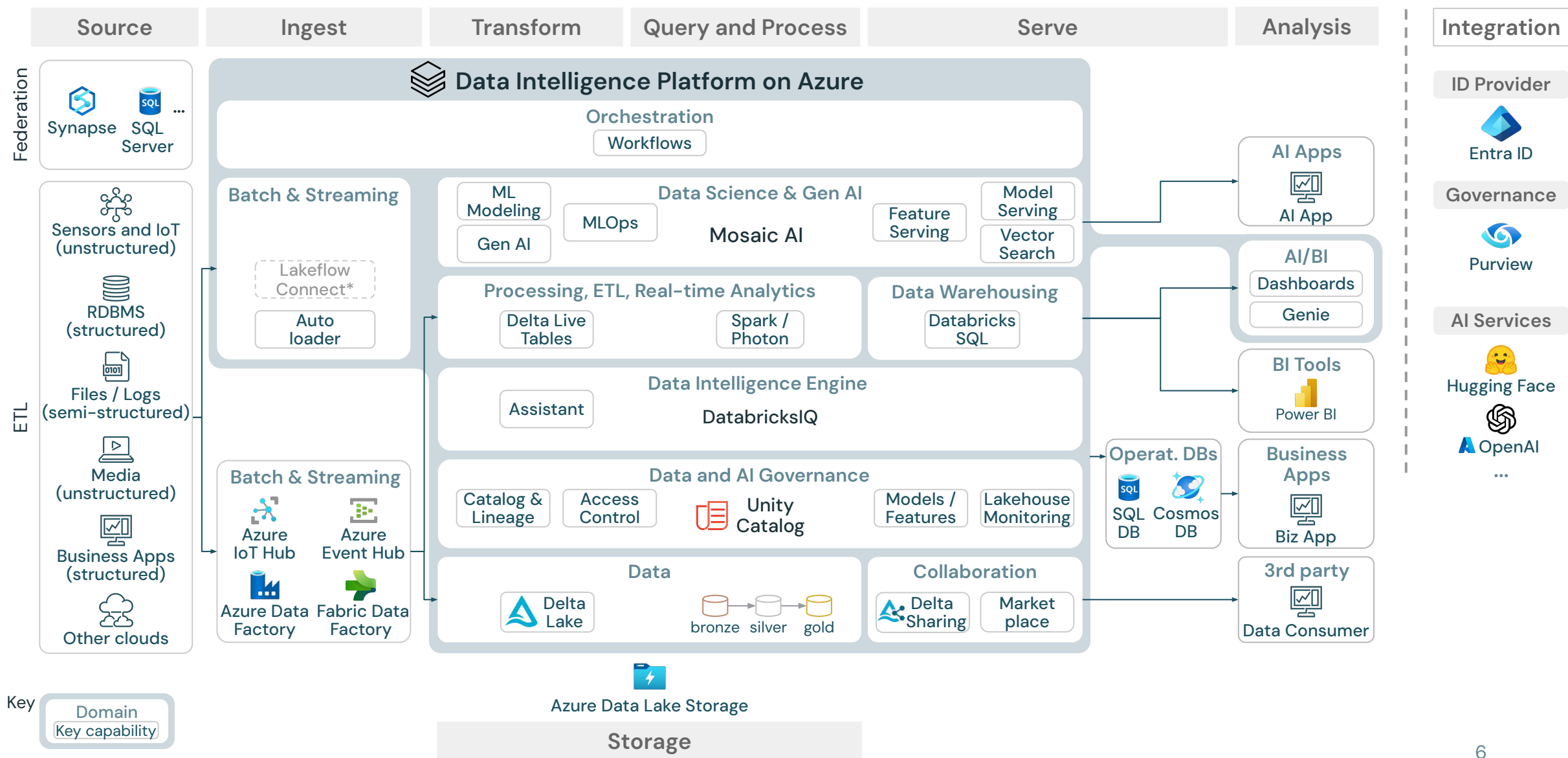
ICEBERG



Databricks “Data Intelligence Platform”



Databricks Data Intelligence Platform on Azure

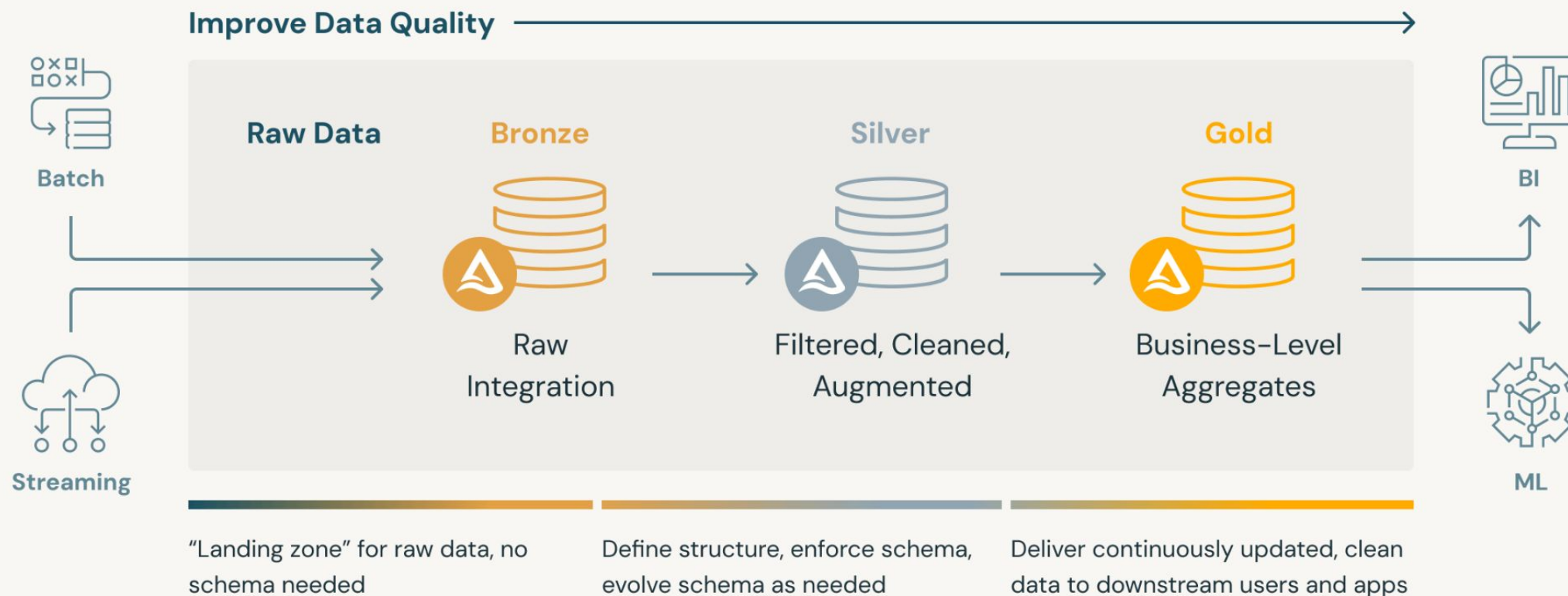


Medallion Architecture



Medallion Architecture – Visualized

Building reliable, performant data pipelines with  **DELTA LAKE**



What is a Medallion Architecture?

- A medallion architecture is a data design pattern used to logically organize data in a lakehouse.
- Goal of incrementally and progressively improving the structure and quality of data as it flows through each layer of the architecture (from Bronze $\Rightarrow$ Silver $\Rightarrow$ Gold layer tables).
- Medallion architectures are sometimes also referred to as "multi-hop" architectures.



Medallion Architecture – Additional Resources

[What is a Medallion Architecture?](#)

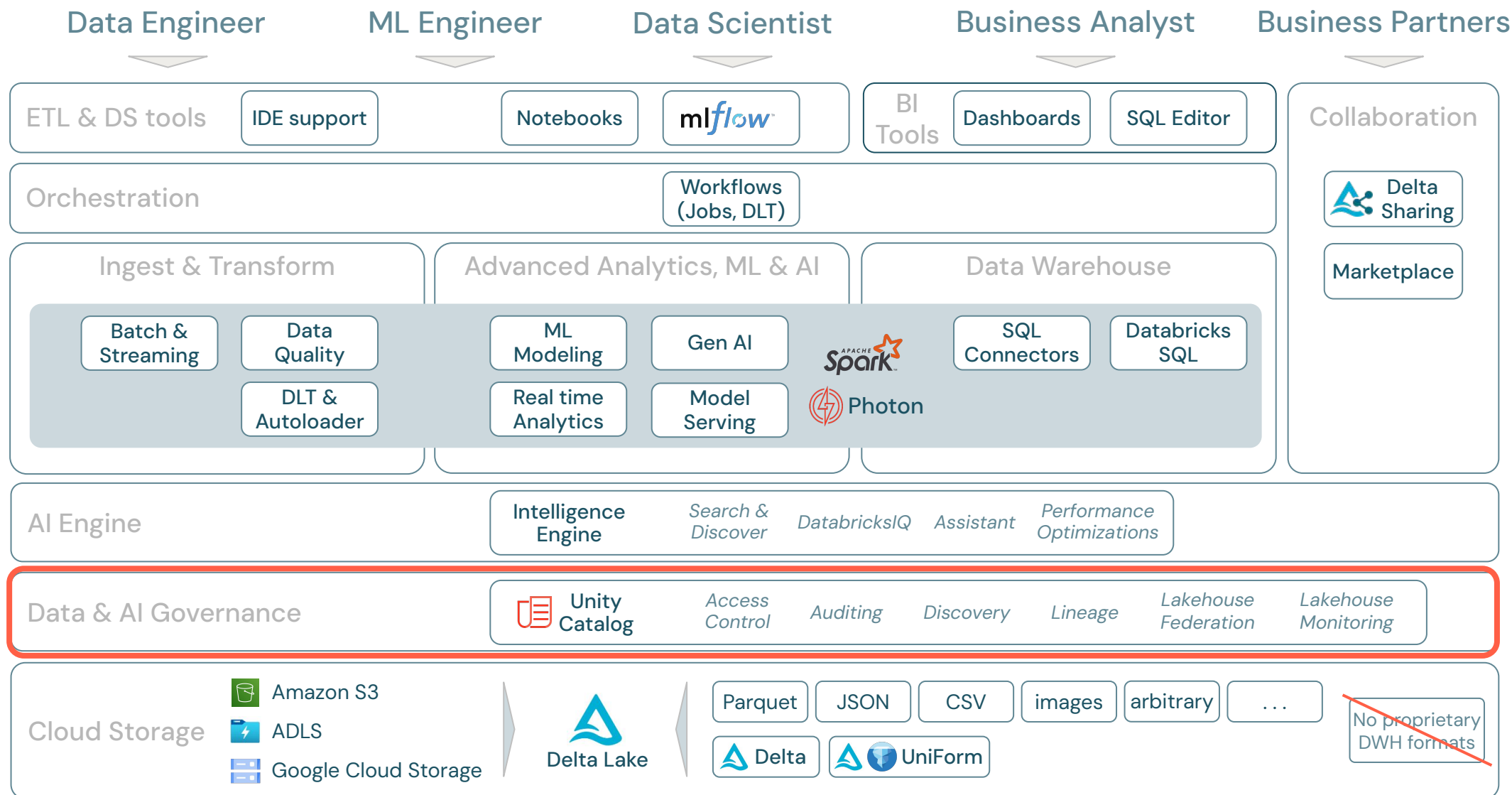
[What is a Data Lakehouse?](#)



Unity Catalog



Databricks “Data Intelligence Platform”



What is Unity Catalog?

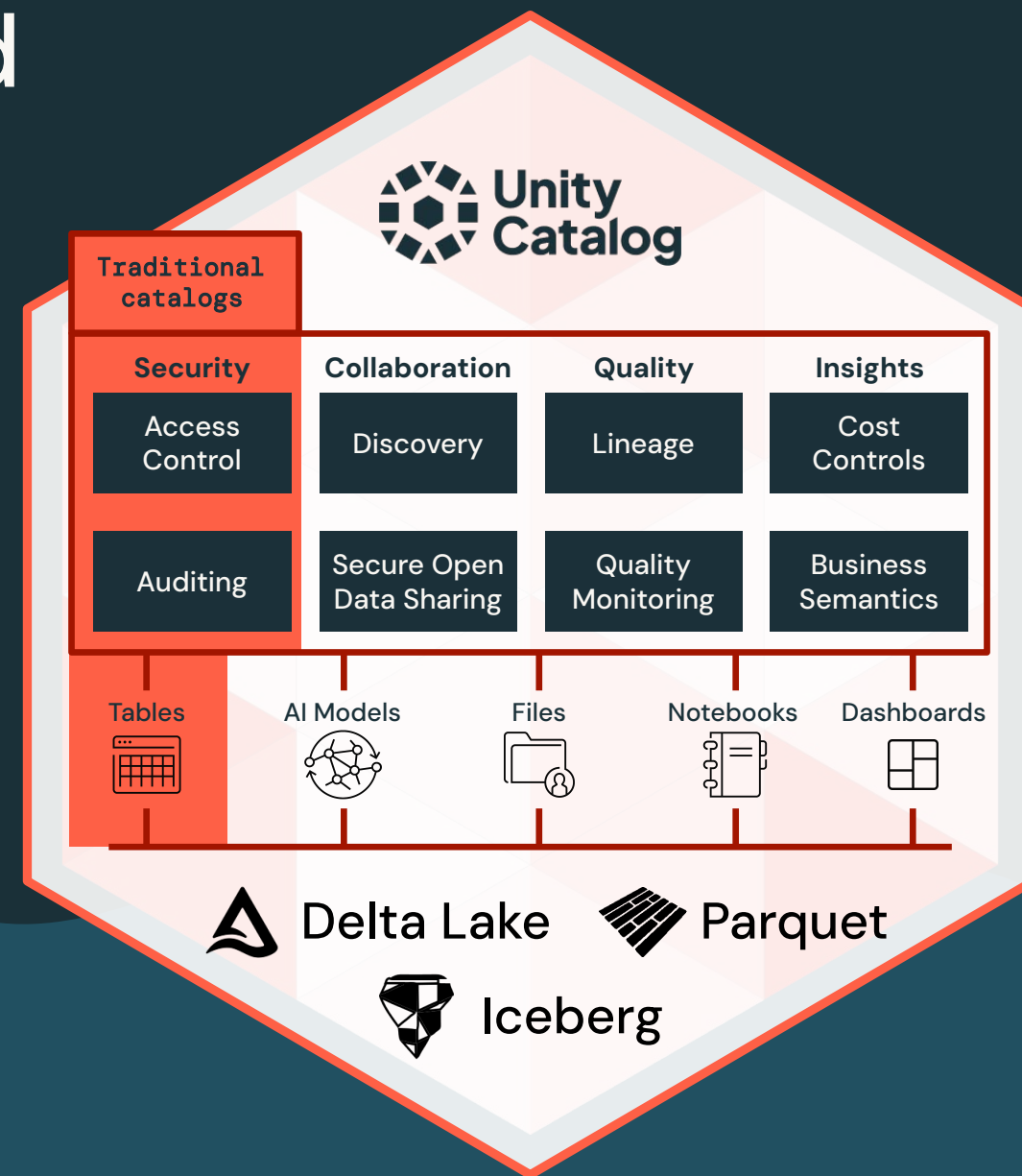
- Unity Catalog provides centralized access control, auditing, lineage, and data discovery capabilities across Databricks workspaces.



Protect and understand your data

Unified capabilities
for every use case

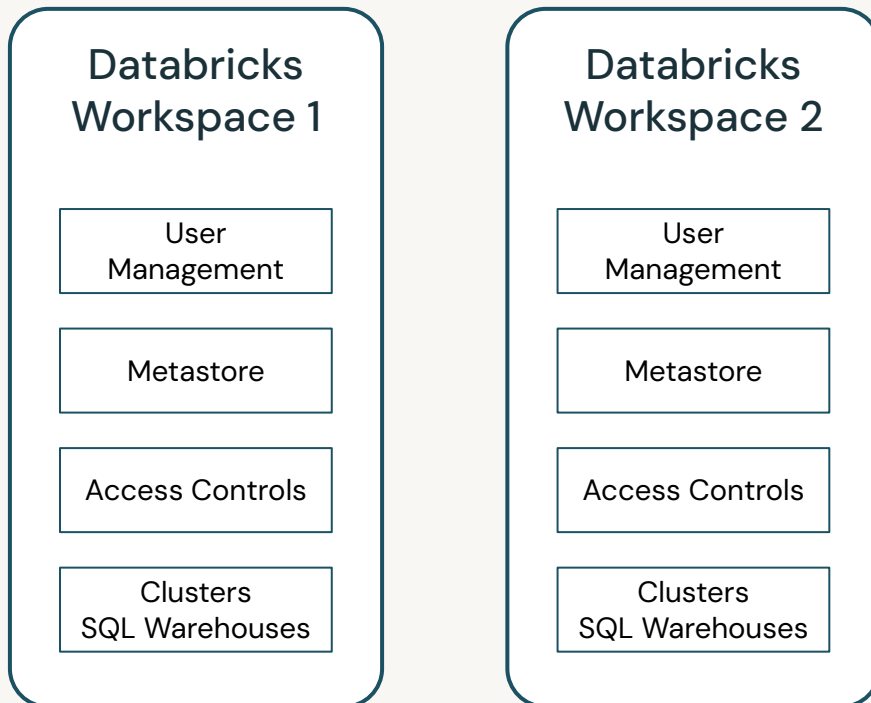
Unified governance
for all assets



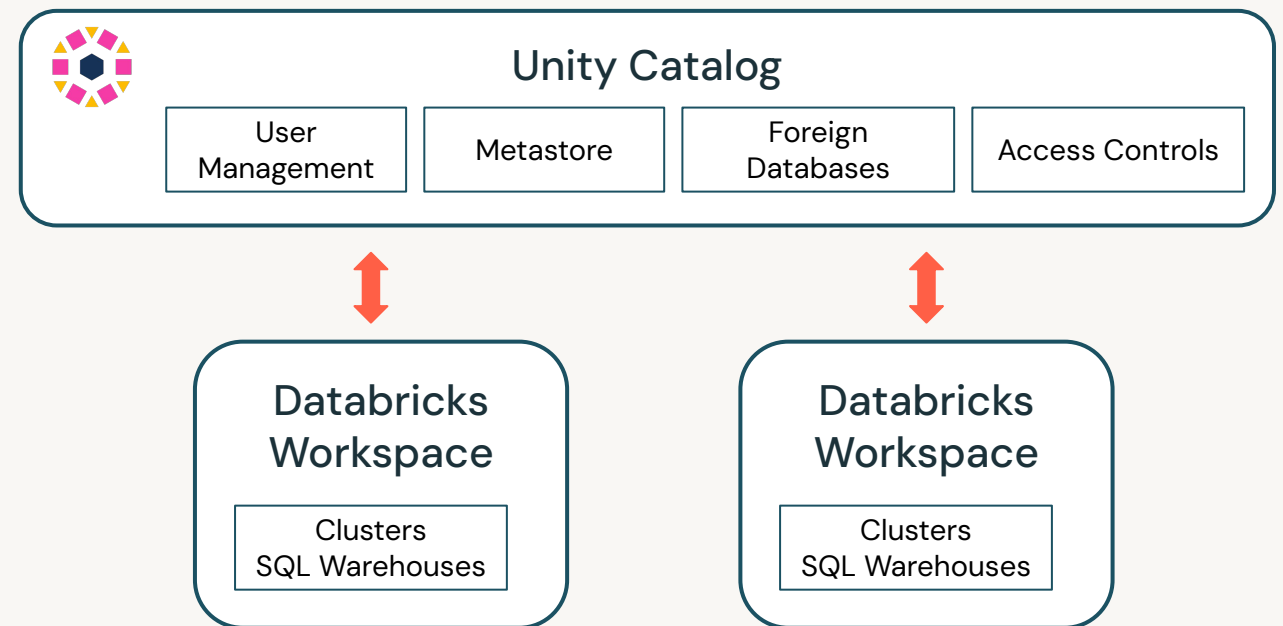
All your metadata, in one place

One metadata layer across file and database sources **superpowers** governance

Without Unity Catalog

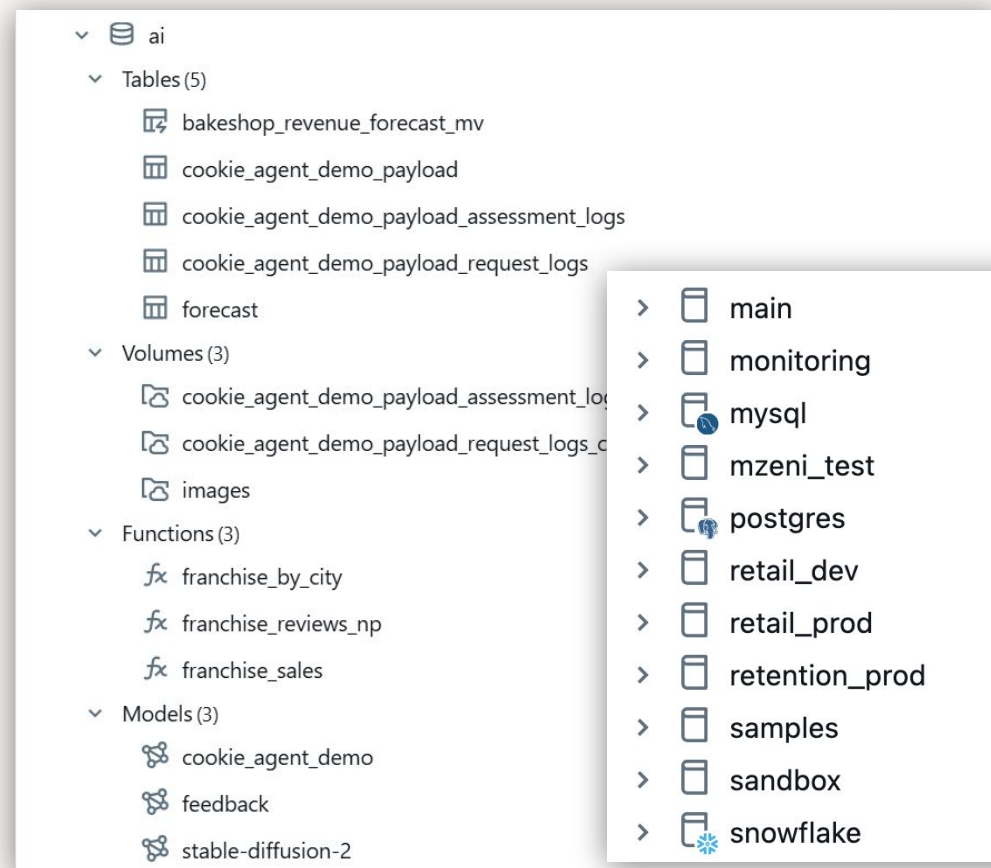


With Unity Catalog



Unified discovery

- Drive operational efficiencies with a single catalog for all files, tables, ML models, AI tools, notebooks, business metrics
- Leverage any open data format of your choice, including Delta, Iceberg, Parquet and more
- Create a single source of truth for all of your data, no matter where it lives



Unified access management

- Improve compliance with simple interface to manage access policies and audit all data and AI
- Enhance data security with fine-grained access controls and attribute based policies
- Accelerate the journey from data discovery to utilization with self-service access workflows

New rule on  sales

General
Enter basic information for your rule

Name
mask ssn

Description
masks ssn where data is tagged with pii:ssn

Subject
Who should this rule apply to?

Applied to...
This rule will affect selected users, groups, or service principals
All account users X

Except for...
Exclude selected users, groups, or service principals from this rule.
For example, a user from a group may be excluded.
Search principals

Type & target
What type of rule is this? Which objects should it apply to

Rule type
☒ Column Mask ☐ Row Filter

Rule target
This rule may be enforced for everything within  sales or

Access request destinations X

Requests for access will be sent to the selected destinations.

Notification destinations

✉ Table owner (you) Default ▾ 🗑

Select a destination ▾ +

#data-requests alerts

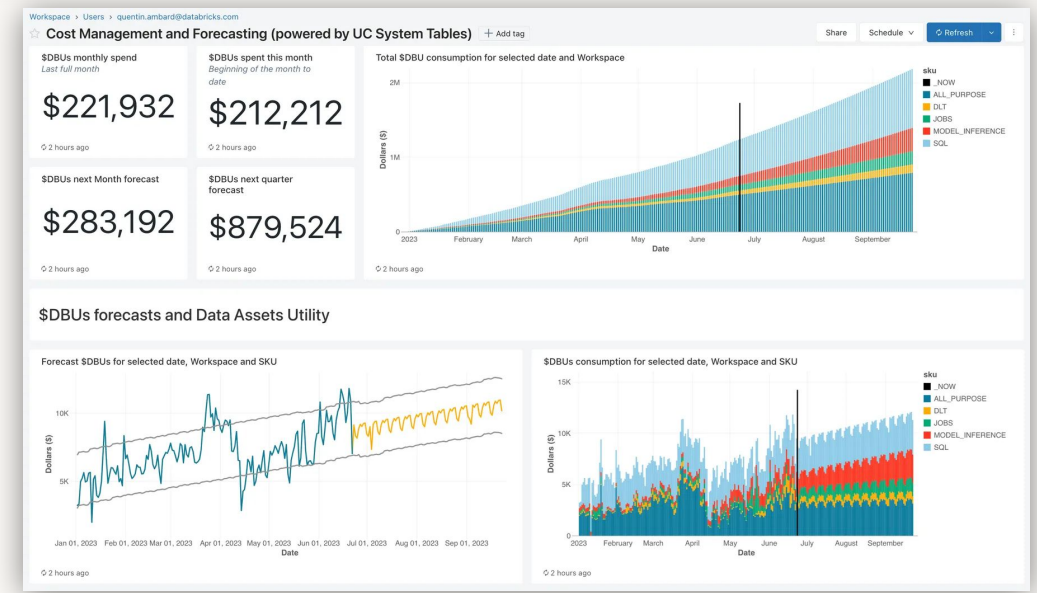
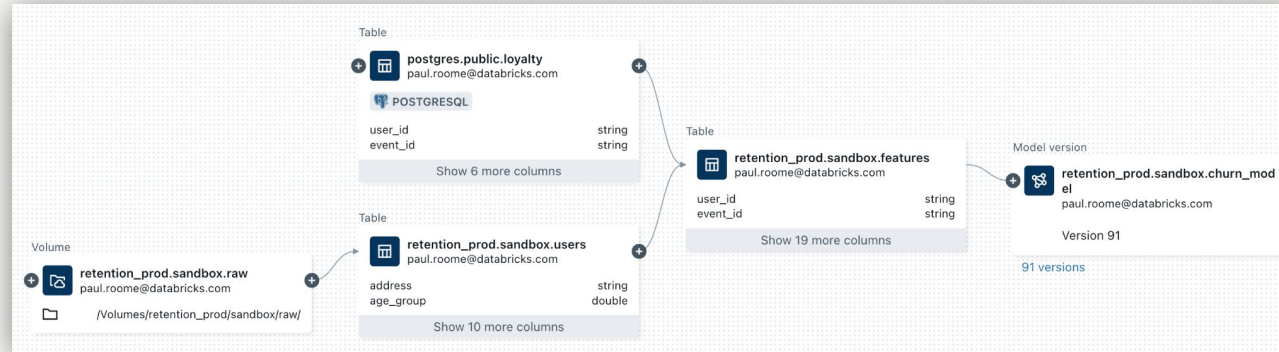
Jira ticket queue

Cancel Update



Unified monitoring and observability

- Reduce operational overhead of tracking data flow and dependencies with real-time, column-level lineage for data & AI
- Improve compliance for sensitive data and AI assets with tagging and auto-classification
- Ensure data and AI quality with AI-powered monitoring and proactive alerting
- Gain end-to-end visibility into billing, usage, cost and compliance



Unity Catalog – Key Features

- Define once, secure everywhere: Unity Catalog offers a single place to administer data access policies that apply across all workspaces.
- Standards-compliant security model: Unity Catalog's security model is based on standard ANSI SQL and allows administrators to grant permissions in their existing data lake using familiar syntax, at the level of catalogs, databases (also called schemas), tables, and views.
- Built-in auditing and lineage: Unity Catalog automatically captures user-level audit logs that record access to your data. Unity Catalog also captures lineage data that tracks how data assets are created and used across all languages.



Unity Catalog – Key Features Cont.

- Data discovery: Unity Catalog lets you tag and document data assets, and provides a search interface to help data consumers find data.
- System tables (Public Preview): Unity Catalog lets you easily access and query your account's operational data, including audit logs, billable usage, and lineage.



Unity Catalog – Additional Resources

[What is Unity Catalog?](#)

[Unity Catalog Best Practices](#)

[Unity Catalog Technical Deep Dive](#) (Video ~ 1.25h)

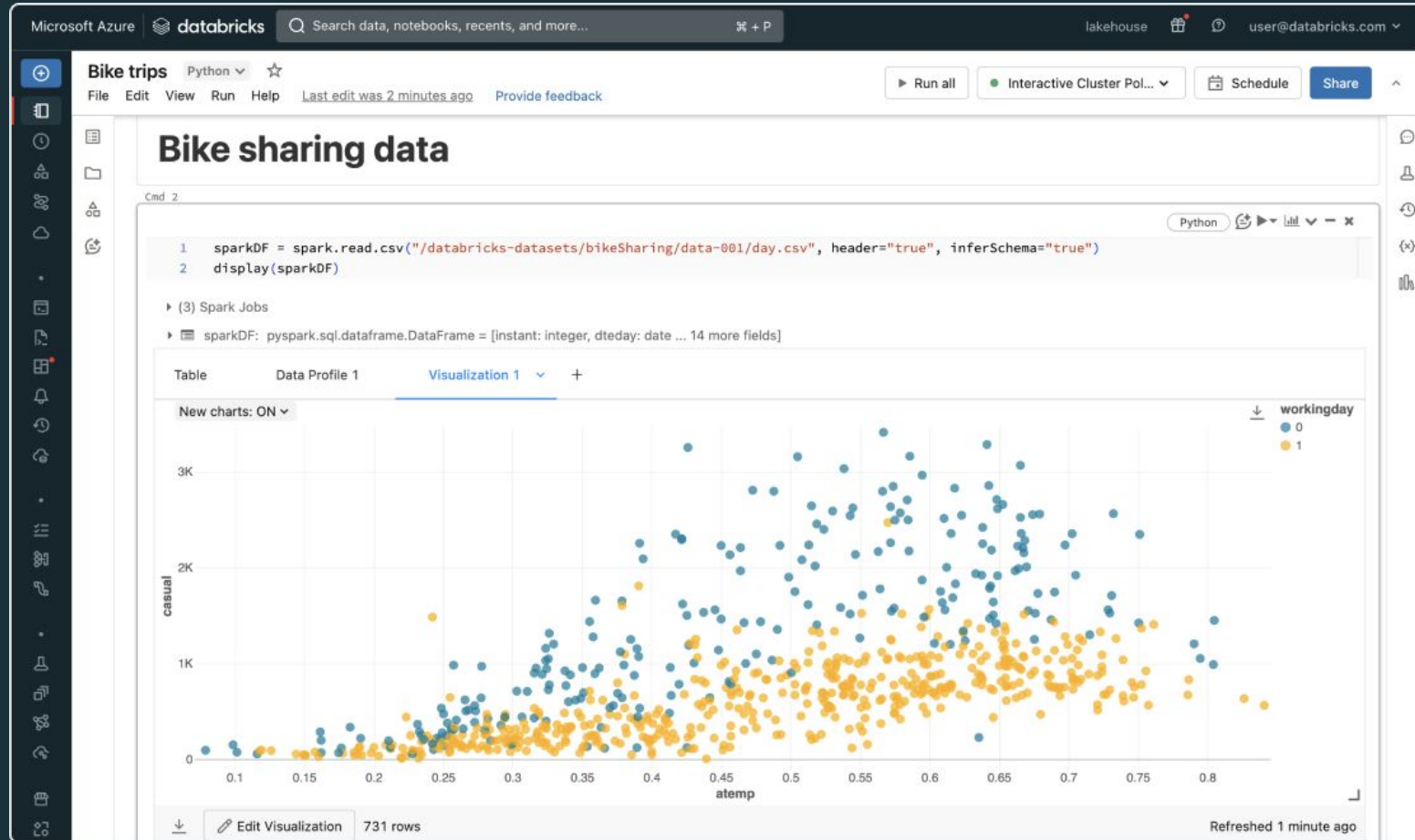


Databricks Notebooks



What are Databricks Notebooks?

- Databricks Notebooks simplify building data and AI projects through a fully managed and highly automated developer experience.
- Notebooks work natively with the Databricks Platform to help data practitioners start quickly, develop with context-aware tools and easily share results.



Databricks Notebooks – Additional Resources

[Introduction to Databricks Notebooks](#)

[Develop Like a Pro in Databricks Notebooks \(~37m Video\)](#)

[Lakehouse for Customer 360 Demo](#)

[Git Operations with Repos](#)

[CI/CD Techniques with Git and Databricks Git Folders](#)



User experience: Ecosystem & ease of use

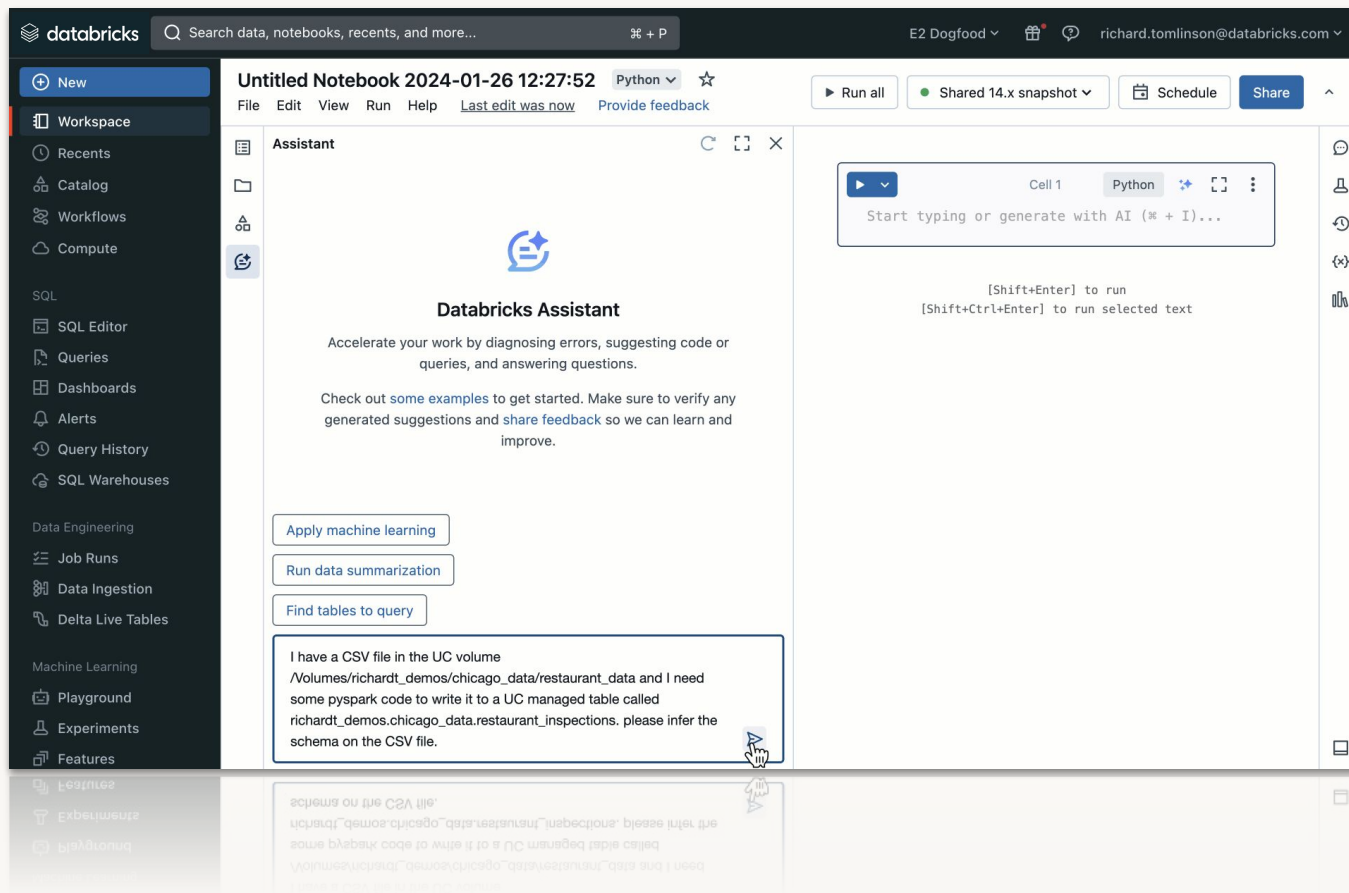


Genie Code

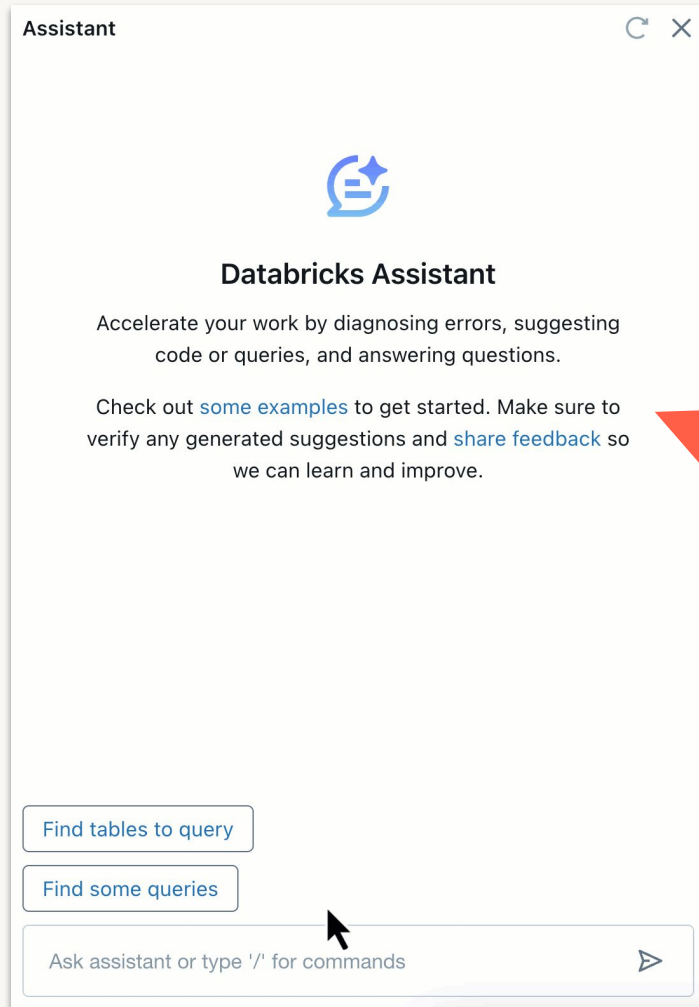
Boost productivity with your context-aware AI assistant



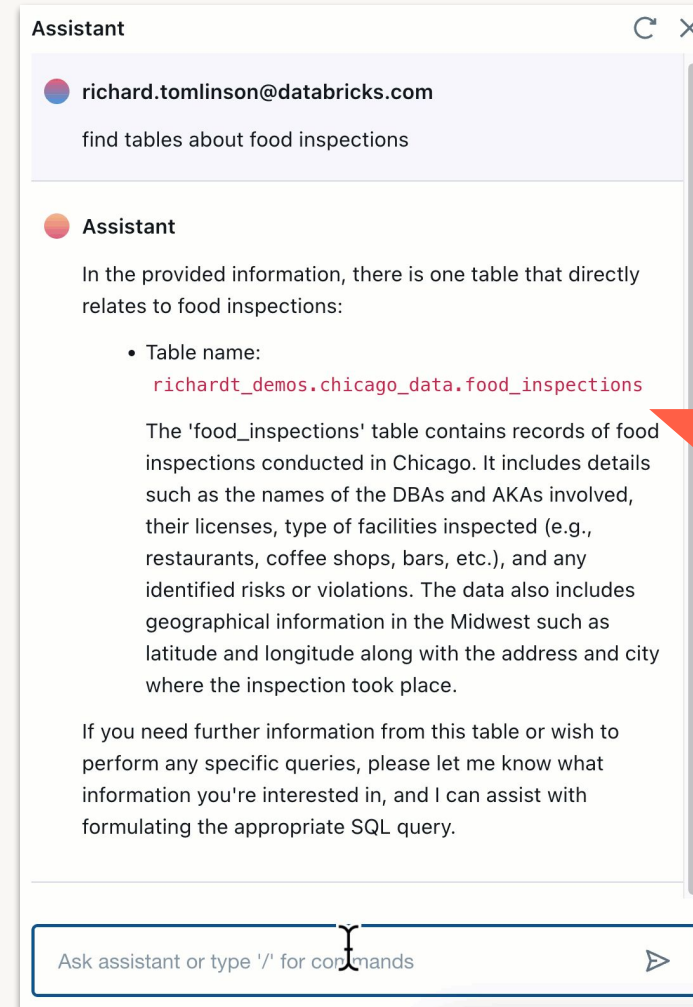
- Generate, explain, and fix code with natural language
- AI assistance in every user experience



Example of a context-aware AI assistant



AI knows where to find the right data and only displays the tables I am allowed to see



AI knows how to create the right SQL relevant to the business question asked

AI assistance in every user experience



Databricks Assistant in Notebooks

Databricks Assistant in SQL editor

Databricks Assistant in Lakeview dashboards

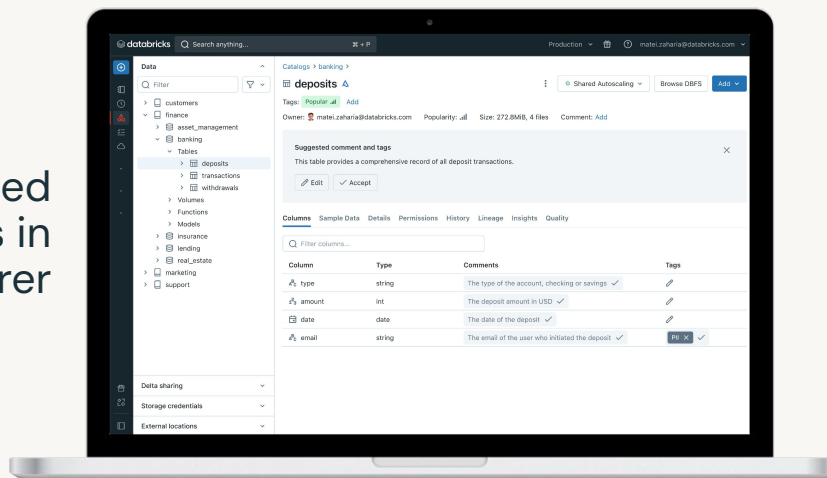
Databricks Assistant in Help

Boosts productivity for all users

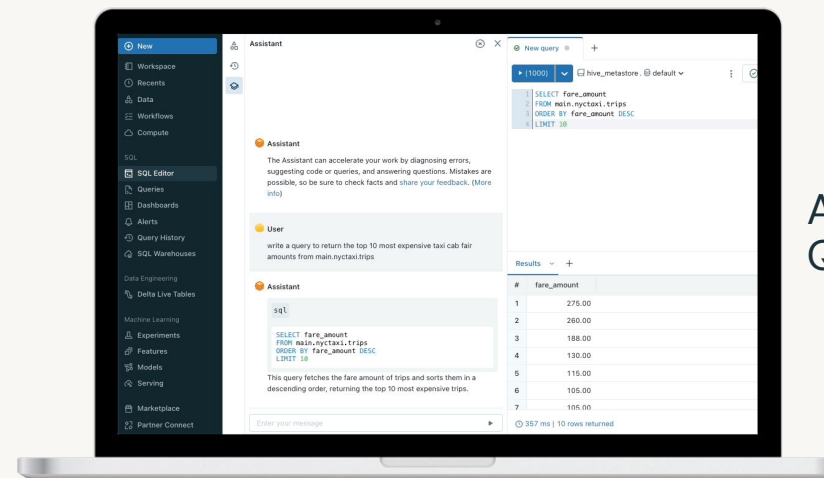
Data Scientists | Data Engineers | SQL Analysts | Business Users

We're infusing AI in our experiences

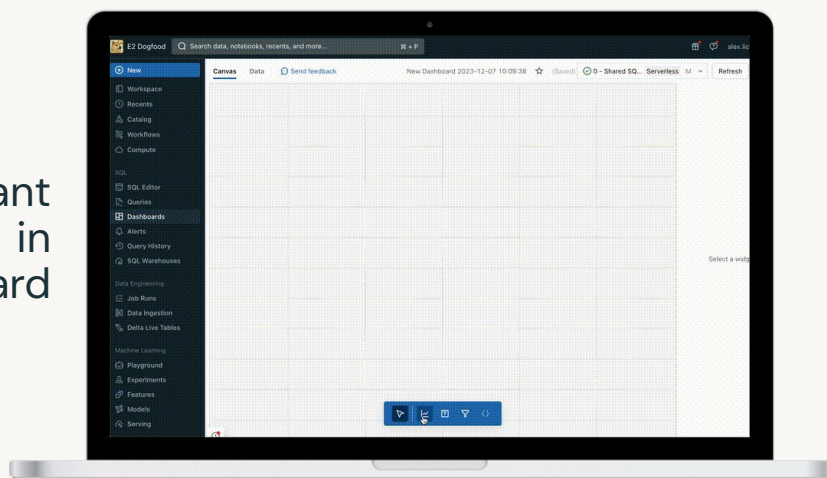
AI-generated
docs in
Catalog Explorer



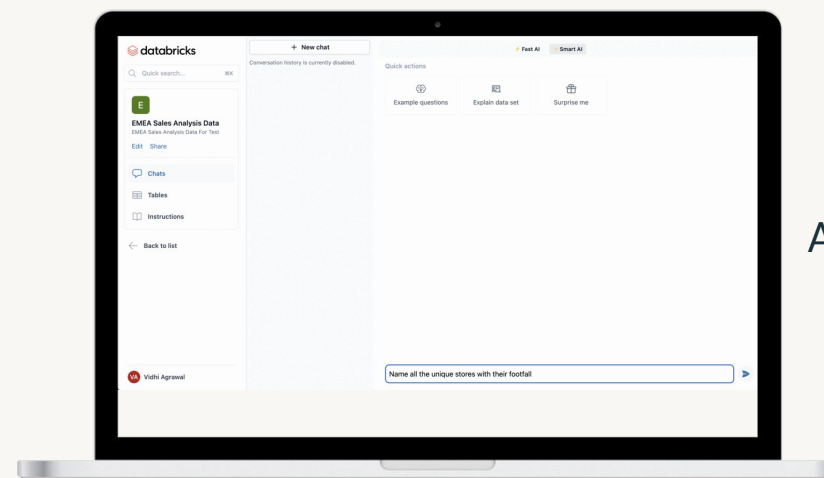
Assistant in
Query Editor



Assistant
in
Dashboard



AI/BI Genie



Choosing Your Compute: Interactive vs. Jobs vs. Serverless

Databricks offers three compute types, each tuned for a different kind of workload.

Interactive

All-Purpose Compute

Shared clusters you start and keep running for hands-on work.

- Notebooks, ad-hoc analysis, team collaboration
- Manual start/stop; stays on until terminated
- Most flexible, but highest cost if left idle

Best for: Development & exploration

Jobs

Jobs Compute

Dedicated clusters created for a scheduled run, then torn down automatically.

- Production ETL, pipelines, automated workflows
- Spun up per run; auto-terminates when done
- Lower DBU rate than all-purpose

Best for: Scheduled production jobs

Serverless

Fully managed by Databricks

Instant, Databricks-managed compute. No clusters to size, configure, or manage.

- SQL, notebooks, jobs, and pipelines
- Starts in seconds, auto-scales to demand
- Pay only for what you use; zero infra ops

Best for: Fast, on-demand workloads

Rule of thumb: explore on Interactive → automate with Jobs → default to Serverless for instant start and no infrastructure to manage.



Serverless: Custom Environments & Dependencies

Yes, you can install custom libraries on serverless. Build your own environment from a Databricks-managed base.

Base Environments

Runtime + pre-installed libraries

Start from a Databricks-managed base that pins your runtime and bundled packages.

- Standard, ML, or AI base images
- Sets your environment version (e.g. Standard v5)
- Admin controls which bases the workspace offers

Best for: A consistent starting point

Add Your Libraries

Custom dependencies on top

Layer your own packages over the base. No cluster config required.

- Notebooks: Environment side pane (requirements.txt)
- Jobs: Configure environment per task
- Sources: public PyPI, workspace files, UC volumes

Best for: Project-specific packages

Reuse & Scope

Workspace-scoped, not account-wide

Export your environment spec and reuse it anywhere.

- Save the spec as YAML (workspace files or UC volume)
- Reference the same YAML across notebooks and jobs
- No account-level global environment

Best for: Reproducible, shareable setups

Rule of thumb: pick a base environment → layer custom libraries (PyPI / workspace files / UC volumes) → export the YAML to reuse it.
Environments are workspace-scoped, not account-wide.



